

Chapter 11

Tilemaps with Tiled

Draw the level. Load the level.



Today

- **Tiled**: tilesets, tile layers, object layers, custom properties
- `.tmx` (source) and `.tmj` (what the game loads)
- GIDs
- `tilemapTiledJSON`, `make.tilemap`, `addTilesetImage`, `createLayer`
- solid tiles **by property**
- reading objects: spawn points, coins, enemies, doors
- the level is **data**, not code

Tiled

level1.tmx - Tiled

File Edit View World Map Layer Help

Stamp Brush (B) Bucket Fill (F) Eraser (E) Select Objects (S) Insert Rectangle (R) Insert Point (I) Insert Polygon (P) 1

Properties 2

Map

Orientation Orthogonal

Width 64

Height 20

Tile Width 32

Tile Height 32

Custom Properties

name Green Hills

+ Add Property

3

Layers 4

Objects object layer

Ground tile layer

Background tile layer

top of the list = drawn on top
Layer > New > Tile Layer / Object Layer

Tilesets 5

platform_tiles

click a tile to paint with it;
drag to pick a block of tiles

Edit Tileset

1 Tools: paint tiles (Stamp, Bucket, Eraser) or make objects (Rectangle, Point, Polygon)

2 Properties: whatever is selected - the map, a layer, a tile or an object - and its custom properties

3 The map itself: 32 x 32 tiles; the grid and the objects are only drawn in the editor

4 Layers: tile layers and object layers, top of the list drawn on top

5 Tilesets: the tiles you paint with

A new map, a new tileset

New Map (File > New > New Map...)

Map

OrientationOrthogonal

Tile layer formatCSV

Tile render orderRight Down

Map size

☒ Fixed☐ Infinite

Width64 tiles

Height20 tiles2048 x 640 pixels

Tile size

Width32 px

Height32 px

Cancel

OK

1

New Tileset (File > New > New Tileset...)

Tileset

Nameplatform_tiles

TypeBased on Tileset Image

☒ Embed in map

Image

Source...tilesets/platform_tiles.png

☐ Use transparent colorBrowse...

Tile width32 px

Tile height32 px

Margin0 px

Spacing0 px

Cancel

OK

2

3

1 a fixed size, in tiles - the map is 64 tiles across and 20 down

2 Embed in map: the tileset is saved INSIDE the map file. Phaser cannot read a separate (external) tileset

3 the name you will pass to addTilesetImage() in Phaser - Tiled suggests the picture's file name

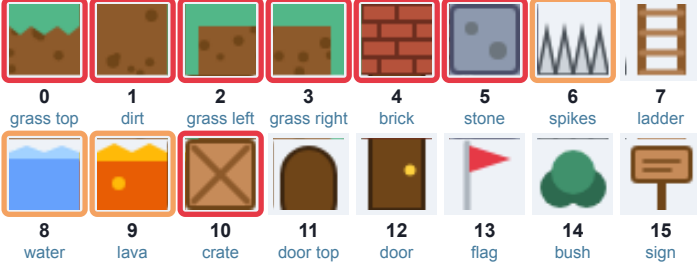
Orthogonal · **fixed** size · CSV · tileset **embedded in map**

Layers, and painting

- the map starts with one tile layer: rename it **Ground**
- Layer > New > Tile Layer: **Background**, below Ground
- top of the Layers list = drawn on top
- Stamp Brush (B), Bucket Fill Tool (F), Eraser (E)
- **check which layer is selected before you paint**

Custom properties on tiles

The tileset, with a custom property on some of its tiles



0 grass top 1 dirt 2 grass left 3 grass right 4 brick 5 stone 6 spikes 7 ladder

8 water 9 lava 10 crate 11 door top 12 door 13 flag 14 bush 15 sign

☐ collides = true (solid: the player stands on these)

☐ hazard = true (touch one and you start again)

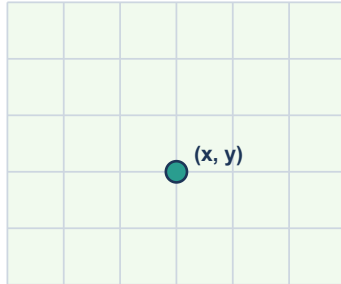
Tiles with no box have no custom properties at all.
The numbers are tile INDEXES within the tileset, from 0 (see the GIDs diagram).

Properties (7 tiles selected)	
Tile	
ID	0
Class	
Probability	1
Custom Properties	
collides	☒ true 1
1 tick it - the value is true	
2 select several tiles first to add it to them all	
+ Add Property: name collides, type bool 2	

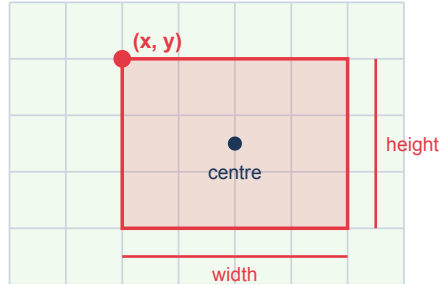
Edit Tileset → select tiles → + → **collides** , **bool**, ticked

Objects

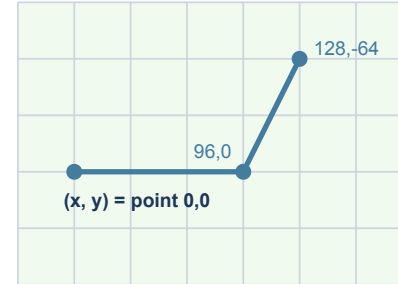
Where Tiled measures each kind of object from



Point: x and y are the point.
A sprite placed there is centred on it.



Rectangle: x, y is the TOP-LEFT corner.
A Zone is placed by its centre:
 $(x + \text{width} / 2, y + \text{height} / 2)$



Polyline: a list of points,
each measured FROM (x, y) .

Layer > New > Object Layer · Insert Point (I) · Insert Rectangle (R) · **Name** them: `player` , `coin` , `enemy` , `goal`

Save, then export

tilde/level1.tmx

File > Save

- you edit this

public/assets/maps/level1.tmx

File > Export As...

- the game loads this

- after the first time: **File > Export** (Ctrl+E / Cmd+E)
- then **build**, then **refresh**
- "I changed the map and nothing happened" = not exported

Inside a .tmj

level1.tmj - the map

```
"type": "map",
"orientation": "orthogonal",
"width": 64, "height": 20,
"tilewidth": 32,
"tileheight": 32,
"infinite": false,
"properties": [ ... ],
"tilesets": [ ... ],
"layers": [ ... ],
"nextlayerid": 4,
"nextobjectid": 21,
"version": "1.10"
```

Keys are written in alphabetical order;
"..." marks what has been left out here.

Phaser reads all of it with
`this.load.tilemapTiledJSON(key, file)`
and builds the map with
`this.make.tilemap({ key })`

tilesets[0] - embedded in the map

```
"name": "platform_tiles",      <- addTilesetImage("platform_tiles", ...)
"firstgid": 1,
"image": "../tilesets/platform_tiles.png",
"columns": 8, "tilecount": 16, "tilewidth": 32, ...
"tiles": [ { "id": 0, "properties": [
  { "name": "collides", "type": "bool", "value": true } ] }, ... ]
```

layers[0] and [1] - tile layers

```
"name": "Ground", "type": "tilelayer",    <- createLayer("Ground", ...)
"width": 64, "height": 20,
"data": [ 0, 0, 0, ...                    one GID per tile, row by row:
          1, 1, 1, 1, 4, 0, 0, 0, 3, ... ] 64 x 20 = 1280 numbers
```

layers[2] - the object layer

```
"name": "Objects", "type": "objectgroup", <- getObjectLayer("Objects")
"objects": [
  { "id": 1, "name": "player", "point": true, "x": 64, "y": 520, ... },
  { "id": 2, "name": "coin", "point": true, "x": 208, "y": 432, ... },
  { "id": 18, "name": "enemy", "point": true, "x": 1232, "y": 528,
    "properties": [ { "name": "speed", "type": "float", "value": 60 } ] },
  { "id": 20, "name": "goal", "x": 1952, "y": 480,
    "width": 32, "height": 64, ... }
]
```

GIDs

Tile index (in the tileset) and GID (in the map)

tileset													
index	0	1	2	3	4	5	6	7	8	9	10	11	...
GID	1	2	3	4	5	6	7	8	9	10	11	12	

GID = firstgid + index. This tileset's firstgid is 1, so GID 0 is left free to mean "no tile".
A second tileset in the same map would start where this one ends - firstgid 17.

row 17 of the Ground layer, columns 7 to 16, as saved in the .tmj:

	1	1	1	1	4	0	0	0	3	1
drawn										
tile.index	1	1	1	1	4	-1	-1	-1	3	1

In Phaser a Tiled tile's index IS its GID - grass top is 1, not 0 - and an empty place is -1.
So `setCollision([1, 2, 3])` means GIDs. Choosing tiles by property avoids counting at all.

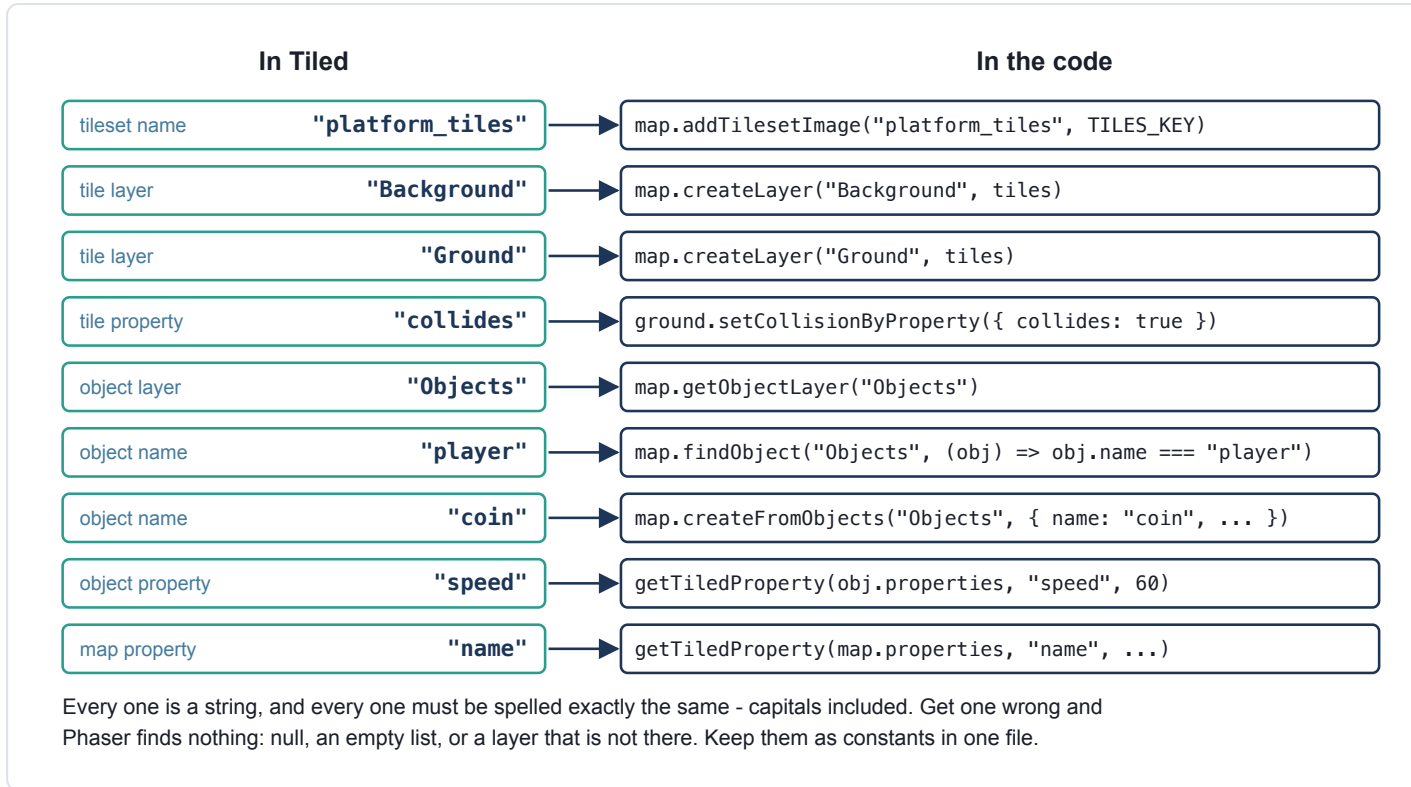
Loading the map

```
// preload()
this.load.tilemapTiledJSON(MAP_KEY, MAP_FILE); // the .tmj
this.load.image(TILES_KEY, TILES_FILE);       // its picture - separately

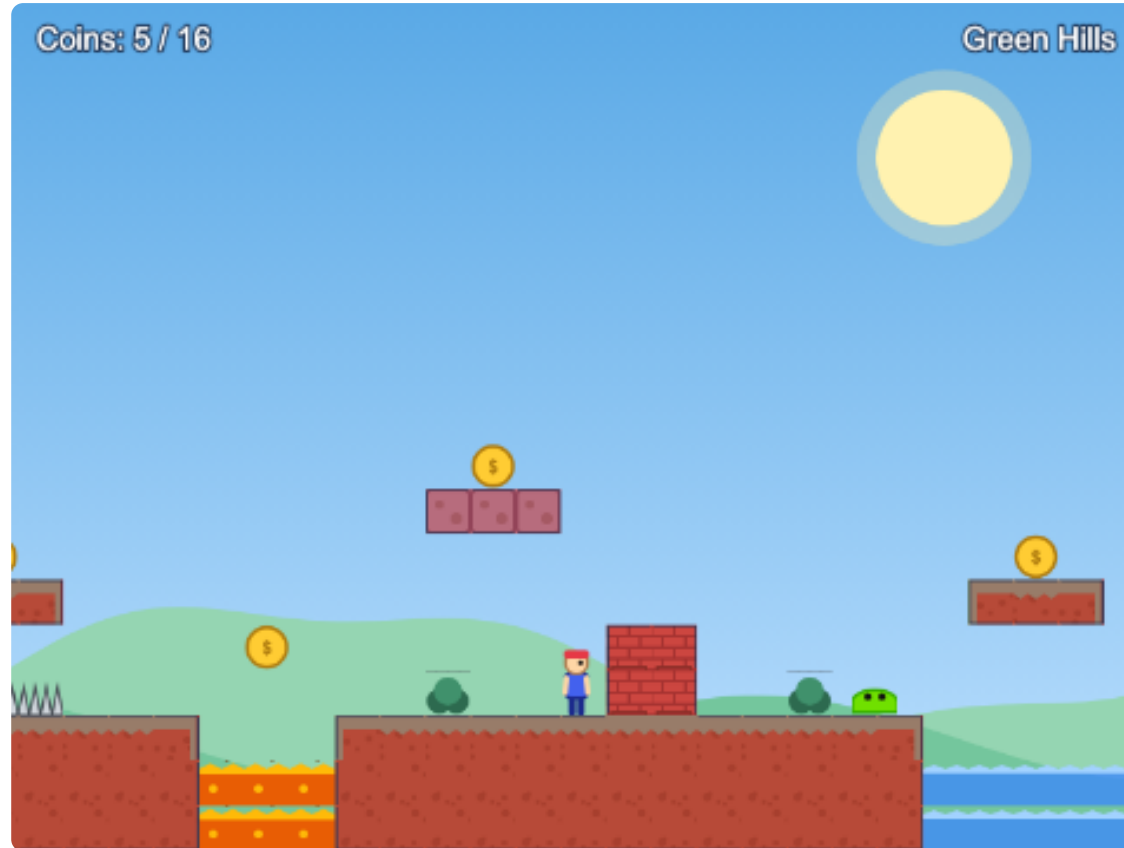
// create()
this.map = this.make.tilemap({ key: MAP_KEY });
const tiles = this.map.addTilesetImage(TILESET_NAME, TILES_KEY);
this.map.createLayer(BACKGROUND_LAYER, tiles);
this.ground = this.map.createLayer(GROUND_LAYER, tiles) as Phaser.Tilemaps.TilemapLayer;
this.ground.setCollisionByProperty({ collides: true });
```

```
addTilesetImage(name in Tiled, key in Phaser)
```

The names must match



Press D



```
tiles.some((tile) => tile.properties.hazard === true); // any tile property
```

Three ways into an object layer

```
// one object
const spawn = this.map.findObject(OBJECTS_LAYER, (obj) => obj.name === "player");

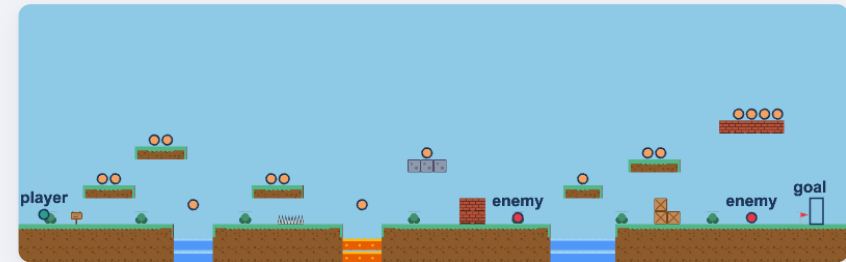
// a Sprite for every match
const coinSprites = this.map.createFromObjects(OBJECTS_LAYER, { name: "coin", key: COIN_KEY });

// all of them, to do as you like with
for (const obj of objectLayer.objects) {
  if (obj.name === "enemy") {
    const speed = getTiledProperty(obj.properties, "speed", DEFAULT_SLIME_SPEED);
```

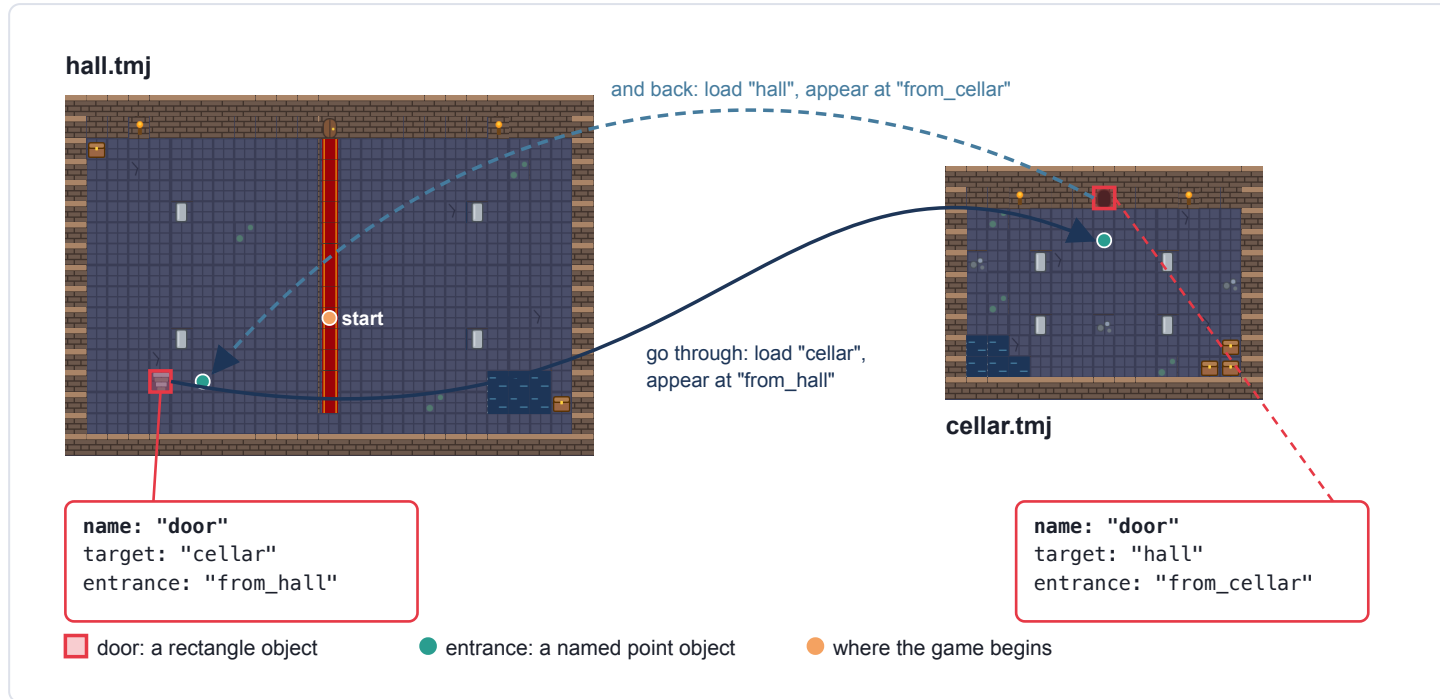
Object and map properties: a list of { name, type, value }

Try it: Tiled Level

- build, serve, play `ch11_tiled_level`
- open `tiled/level1.tmx` in Tiled
- change a slime's `speed` ; add a platform and two coins
- export, build, refresh



Doors between maps



One scene, restarted

```
init(data: Partial<RoomData>): void {  
  this.room = {  
    map: data.map ?? FIRST_ROOM.map,  
    entrance: data.entrance ?? FIRST_ROOM.entrance,  
  };  
  this.leaving = false;  
}  
  
// preload(): only this room's map - skipped if already in the cache  
this.load.tilemapTiledJSON(this.room.map, MAPS_FOLDER + this.room.map + MAP_EXTENSION);  
  
// goThrough(door), after the fade:  
const next: RoomData = { map: door.target, entrance: door.entrance };  
this.scene.restart(next);
```

Try it: Tiled Doors

- down the stairs; back through the cellar door
- where is the list of rooms? (there isn't one)
- why are the entrance points away from the doors?



Summary

- Tiled: tilesets, tile layers, object layers, custom properties
- edit `.tmx`, **export** `.tmj`; embedded tileset, fixed size, CSV
- GID = firstgid + index; in Phaser, `tile.index` is the GID, empty = -1
- `tilemapTiledJSON` → `make.tilemap({ key })` → `addTilesetImage` → `createLayer`
- `setCollisionByProperty({ collides: true });` `tile.properties`
- `findObject`, `filterObjects`, `createFromObjects`, `getObjectLayer(...).objects`
- names are the contract: constants, spelled exactly

Challenges

1. **Above the player** - a layer drawn over the player
2. **Gems** - coins with a `value` property
3. **A third room** - behind the closed door (how much code?)
4. **Signposts** - rectangles with messages
5. **Moving platforms** - along a polyline
6. **Spinning coins, as tiles** - a second tileset, animated in Tiled

Next: **Chapter 12 - Higher or lower**